Lecture 7: Planetary Surfaces
Geology, Geomorphology, & Geophysics

Planetary Systems

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Olympus Mons, Mars. Credit: NASA/JPL/USGS

Learning objectives

By the end of this lecture, you will be able to:

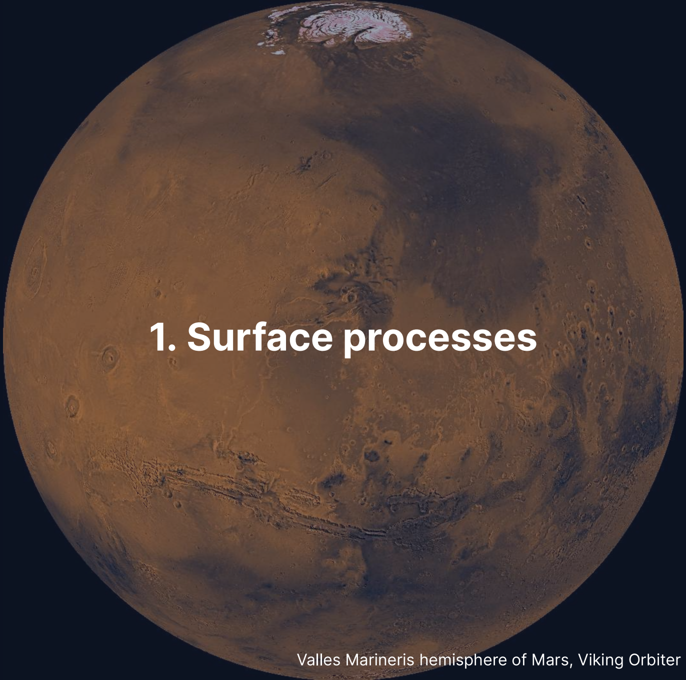
- Distinguish endogenic and exogenic surface processes
- Describe the three stages of impact crater formation
- Derive the crater scaling law $D \sim (E/\rho g)^{1/4}$ from dimensional analysis
- Use crater counting to determine relative and absolute surface ages
- Compare volcanic, tectonic, and erosional styles across the solar system

Recap: Lectures 5 & 6

- Hydrostatic equilibrium and the scale height set an atmosphere's structure
- Greenhouse warming: $T_s > T_{\mathrm{eff}}$; Clausius-Clapeyron sets where clouds form
- The carbonate-silicate cycle is Earth's long-term thermostat

Today: What shapes the *surfaces* of planets and moons?

1. Surface processes



Valles Marineris hemisphere of Mars, Viking Orbiter mosaic. Credit: NASA/JPL/USGS

Endogenic vs. exogenic processes

A surface is a geological record: processes that build landforms compete with processes that destroy them.

Endogenic (internal heat):

- Volcanism
- Tectonics (faulting, folding)

Exogenic (external forces):

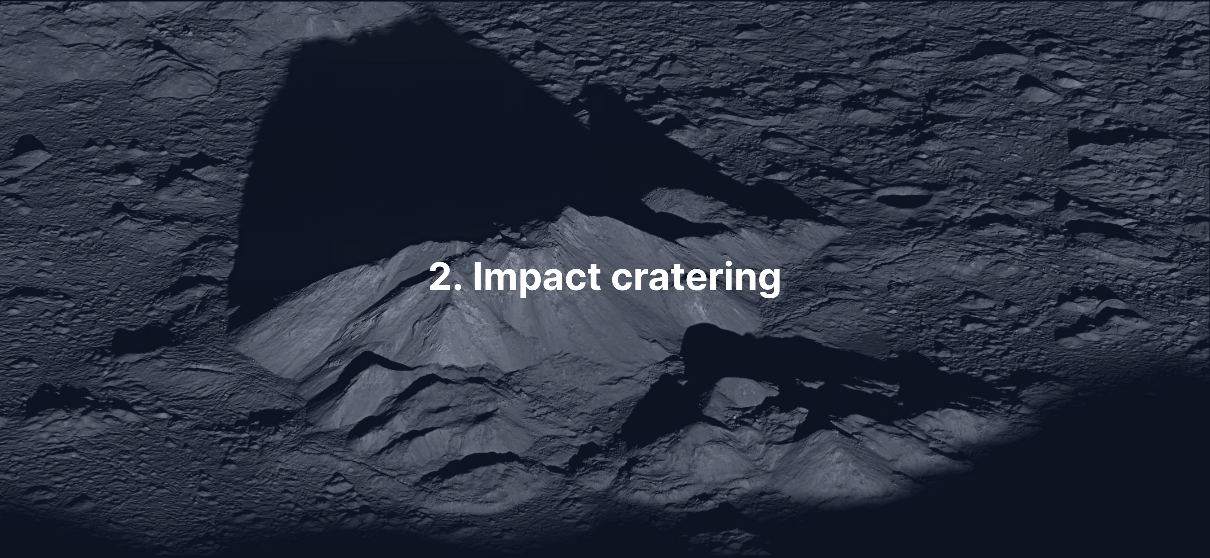
- Impact cratering
- Erosion (wind, water, ice)

Hot interior: endogenic processes win. Dead interior: the surface keeps its ancient craters.

Dominant surface process by body

Body	Dominant process	Key evidence
Moon	Impact cratering	Saturated highland craters, >4 Gyr
Earth	Plate tectonics + erosion	Ocean floors <200 Myr; rapid weathering
Mars	Volcanism + impacts	Olympus Mons; cratered southern highlands
Venus	Volcanism	Volcanic plains cover ~80% of surface
Io	Tidal volcanism	~300–400 active volcanoes; age <1 Myr
Europa	Ice tectonics	Lineae, chaos terrain, very few craters

2. Impact cratering



Lunar crater morphology. Credit: NASA/GSFC/ASU

The most universal geological process

- **Every** solid body in the solar system bears impact craters
- Without atmosphere or geology (Moon, Mercury), craters survive billions of years
- Impact velocities 10–70 km s^{-1}, set by orbital mechanics (Lecture 2)

A 1 km rocky asteroid at 20 km s^{-1} carries $E_k = \frac{1}{2}mv^2 \approx 3 \times 10^{20}$ J, about 1000 times the largest nuclear weapon, released in under a second.

Three stages of crater formation

1. Contact and compression

- Shock wave into target and impactor
- ~100 GPa; melting and vaporisation
- Over in < 1 s

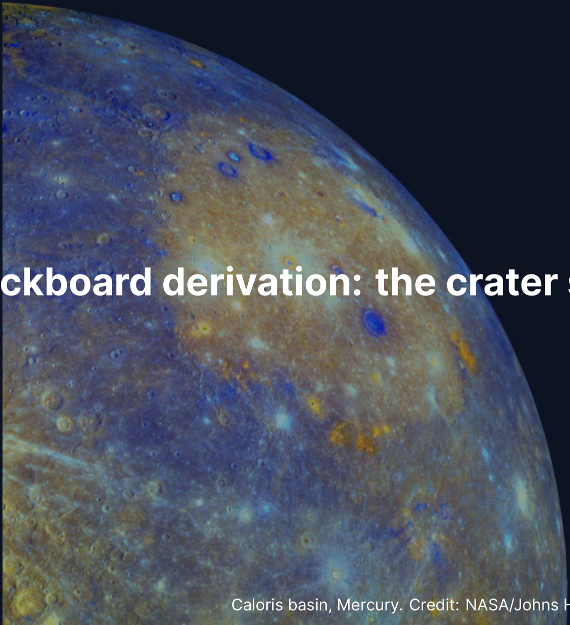
2. Excavation

- Rarefaction wave throws material out
- Transient cavity, depth:diameter ≈ 1:3
- Ejecta blanket around the rim

3. Modification

- Cavity collapses under gravity
- Small: walls slump, **simple crater**
- Large: floor rebounds, **central peak**

3. Blackboard derivation: the crater scaling law



Caloris basin, Mercury. Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington

Setup: crater size from dimensional analysis

Goal: How does crater diameter D depend on impact energy E, target density ρ, and surface gravity g?

In the **gravity regime** (craters ≳100 m, where gravity limits size):

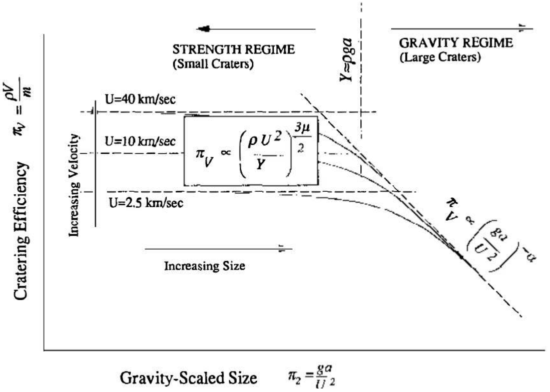
$$D = C\, E^a\, \rho^b\, g^c$$

Dimensions (M, L, T):

$$\begin{aligned}
[D] &= L \\
[E] &= ML^2T^{-2} \\
[\rho] &= ML^{-3} \\
[g] &= LT^{-2}
\end{aligned}$$

To the board.

Where the fourth-root law applies

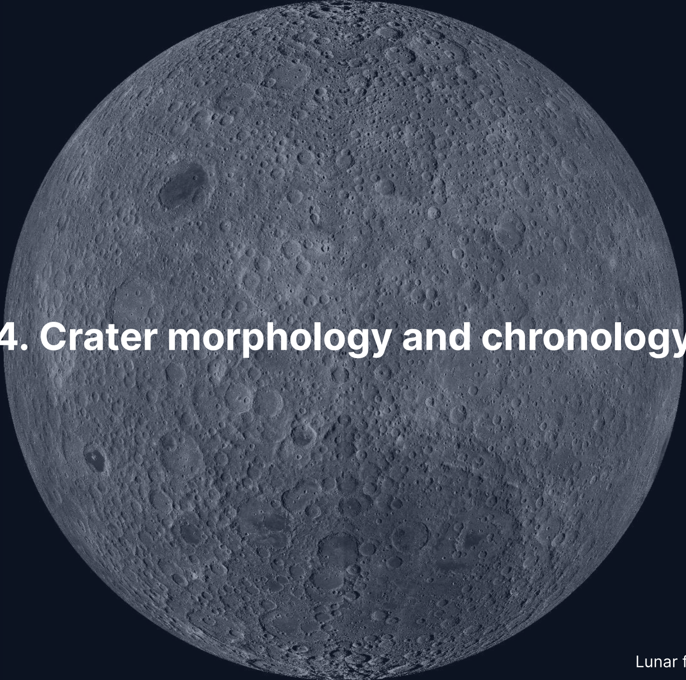


Credit: Holsapple (1993), Fig. 3; curves for U = 2.5, 10 and 40 km s^{-1}

$D \sim (E/\rho g)^{1/4}$: doubling E grows D by only $2^{1/4} \approx 1.19$

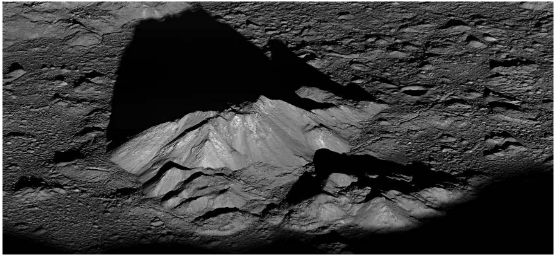
- **Left, strength regime:** cohesion limits growth, curves separate
- **Right, gravity regime:** one law; planetary craters live here; below ~100 m on the Moon the law fails
- At fixed v and ρ: $E \propto L^3$, so $D \propto L^{3/4}$ (1 km impactor, 16.5 km crater)

4. Crater morphology and chronology



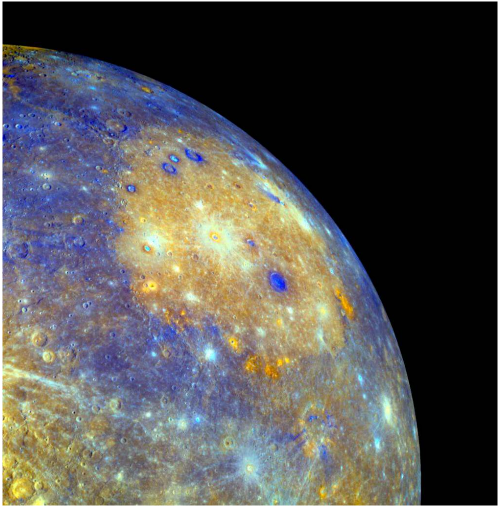
Lunar farside. Credit: NASA/GSFC/ASU

Crater morphology: three classes



Credit: NASA/GSFC/ASU

1. **Simple** ($D < D_t$)
   Bowl-shaped; depth:diameter ≈ 1:5
   Moon: $D_t \approx 15$ km
2. **Complex** ($D_t < D \lesssim 300$ km)
   Central peak, terraced walls, flat floor: gravity overcomes wall strength
3. **Multi-ring basins** ($D \gtrsim 300$ km)
   Orientale (930 km), Caloris (1550 km), Hellas (2300 km)



Caloris basin, Mercury, ~1550 km in diameter: concentric rings, and volcanic plains (orange) that filled it after the impact. Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington.

Transition diameter

The simple-to-complex transition scales with surface gravity:

$$D_t \approx D_{t,\mathrm{Moon}} \cdot \frac{g_{\mathrm{Moon}}}{g}$$

Body	g **(m s^{-2})**	D_t **(km)**
Moon	1.62	15
Mars	3.71	~6.5
Earth	9.81	~2.5
Mercury	3.70	~6.6

On Earth, virtually all craters larger than a few km are complex.

Crater counting: older surfaces carry more craters

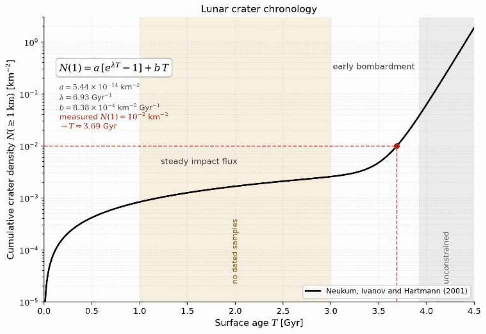
The cumulative size-frequency distribution is a power law:

$$N(>D) = a\,D^{-b}, \quad b \approx 2\text{–}3$$

- $N(>D)$: craters per unit area larger than D; a straight line on log-log axes
- Higher line = older surface: crater counting gives **relative** ages, until **saturation**, when new craters erase old ones
- Apollo and Luna samples date specific surfaces: the **absolute** calibration

Source: Neukum et al. (2001), *Space Sci. Rev.*

The lunar chronology



Credit: Course figure, after Neukum, Ivanov and Hartmann (2001)

$$N(1) = a\left[e^{\lambda T} - 1\right] + b\,T$$

- Linear term: steady flux of the last ~3 Gyr; exponential term: the early bombardment before ~3.6 Gyr
- Read it backwards: $N(1) = 10^{-2}$ km^{-2} gives ~3.7 Gyr
- Highlands saturated, >4 Gyr; maria 3.9–3.1 Gyr; extrapolated to Mars and Venus

The shaded bands hold no dated sample. There the curve is interpolation, not calibration.

5. Volcanism

Olympus Mons, Mars. Credit: NASA/JPL/USGS

Effusive vs. explosive volcanism

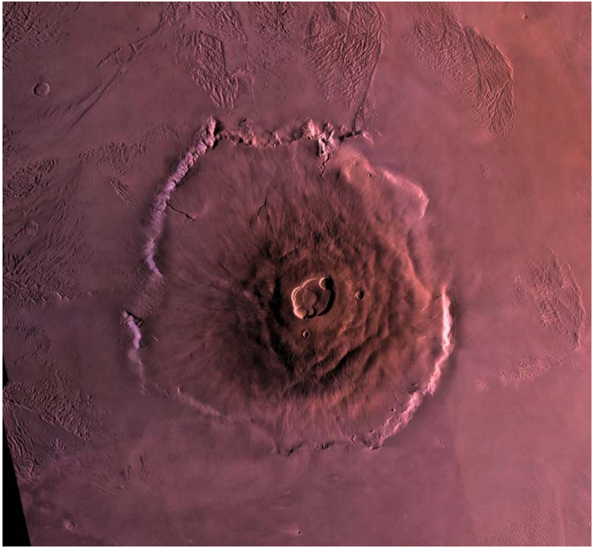
The key variable is **magma viscosity**, set by the $\mathrm{SiO_2}$ content:

Basaltic, ~50% $\mathrm{SiO_2}$**:**

- Flows easily, volatiles escape
- **Effusive**: shields, flood basalts
- Moon, Mars, Io

Silicic, >65% $\mathrm{SiO_2}$**:**

- Traps volatiles until failure
- **Explosive**: stratovolcanoes, ash
- Primarily Earth



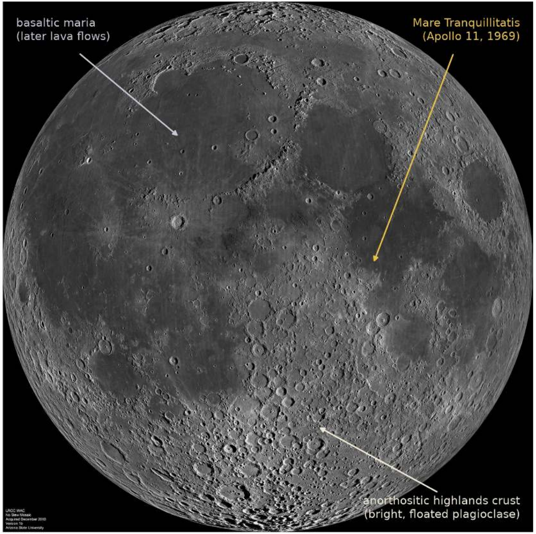
Olympus Mons, Mars: a shield volcano ~600 km wide and ~21 km high, with nested summit calderas. Viking Orbiter mosaic. Credit: NASA/JPL/USGS.

Why is Olympus Mons so large?

Credit: Course figure

Heights on one linear scale, widths schematic: Olympus Mons 21.2 km, Mauna Kea 10.2 km from the sea floor, Everest 8.8 km.

No plate motion: one plume feeds one edifice for billions of years. Lower gravity, $g \approx 0.38\, g_{\oplus}$, lets the crust hold a ~2.6× taller pile.



The lunar maria: flood basalts that filled the nearside basins between 3.9 and 3.1 Ga and cover ~16% of the surface. LROC wide-angle mosaic. Credit: NASA/GSFC/Arizona State University.

Volcanism across the solar system

Body	Style	Driving mechanism	Examples
Earth	Effusive + explosive	Internal heat + recycling	Hawaii, mid-ocean ridges
Moon	Flood basalts (extinct)	Residual heat (3.9–3.1 Ga)	Maria
Mars	Shield volcanoes	Mantle plumes, no plates	Olympus Mons, Tharsis
Venus	Lava plains + shields	Internal heat, stagnant lid	Maat Mons
Io	Ultramafic flows	Tidal heating	Loki Patera, Pele

Venus returns in Lecture 9, Io in Lecture 11.

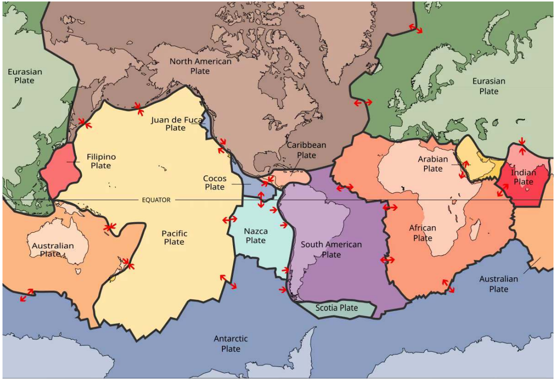
Break

Earthrise, Apollo 8 (1968). Credit: NASA

6. Tectonics

The San Andreas Fault across the Carrizo Plain, California. Credit: Ikluft, Wikimedia Commons, CC BY-SA 4.0

Plate tectonics: Earth's unique regime



Credit: Wikimedia, CC BY-SA 3.0

Earth is the **only** body with active plate tectonics:

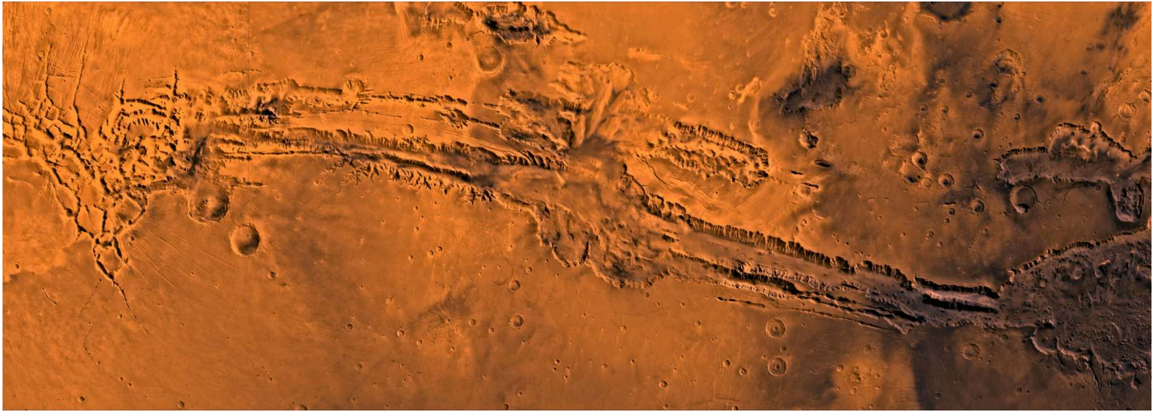
- ~15 rigid plates, moving at 1–10 cm yr^{-1}
- **Divergent:** ridges, new crust
- **Convergent:** subduction, recycling
- **Transform:** horizontal sliding

Subduction closes the **carbonate-silicate cycle**.

Stagnant-lid convection

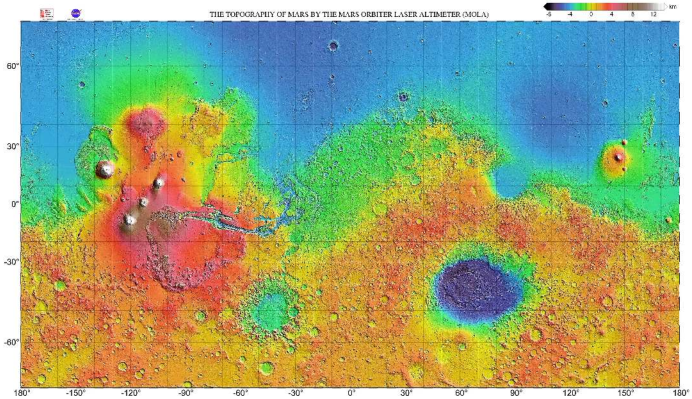
- Every other terrestrial body: one rigid lithospheric lid
- The mantle convects beneath it; heat leaves by conduction and occasional volcanism
- The thick lid insulates: the mantle stays warm, but melt reaches the surface only through rare plumes (Tharsis), and volcanism fades as the lid grows

Stagnant lid is the default for temperature-dependent viscosity. Breaking the lid, as Earth does, needs water weakening and self-sustaining damage.



Valles Marineris in a Viking Orbiter mosaic, ~4000 km long and up to 7 km deep: no plate boundaries, yet the largest rift in the solar system, opened by the Tharsis volcanic load. Credit: NASA/JPL/USGS.

Mars topography: the hemispheric dichotomy



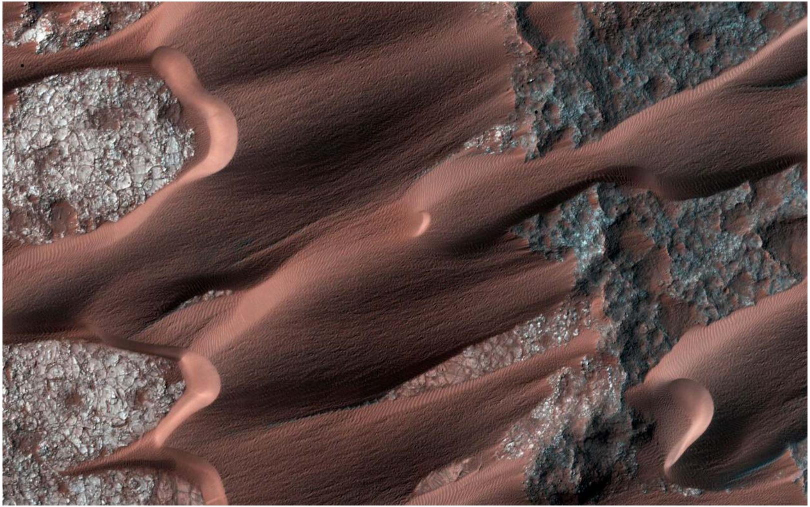
Credit: NASA/JPL/GSFC/MOLA Science Team

~6 km between the southern highlands and the northern lowlands. Tharsis with 4 giant volcanoes; Hellas at −8.2 km; Valles Marineris at the equator.

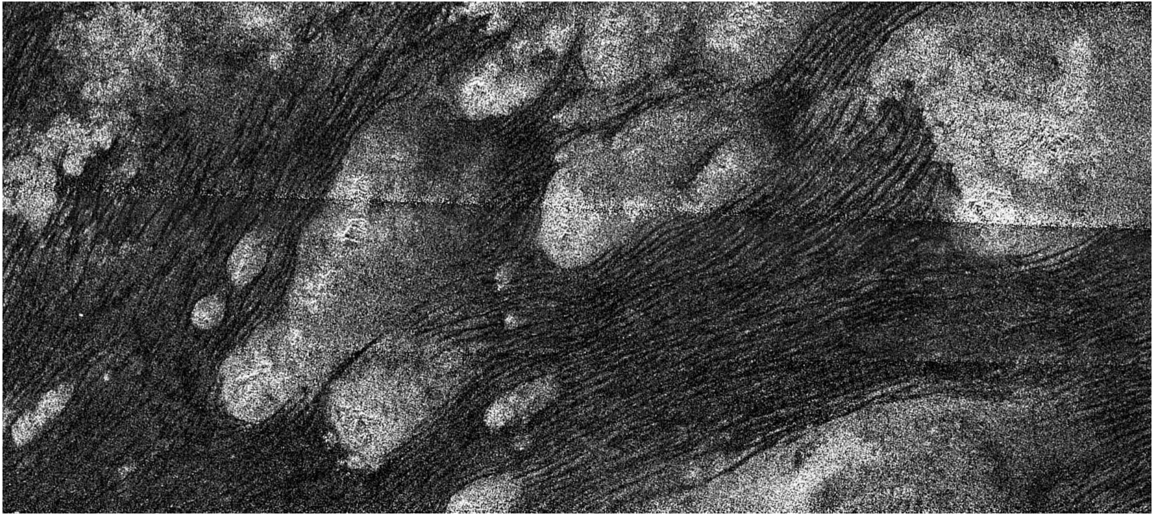
A lobate scarp on Mercury, MESSENGER: a thrust fault that cuts craters, so the contraction is younger than they are; the cooling iron core shrank the radius by up to ~7 km. Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington.

7. Erosion and weathering

Barchan dunes in Nili Patera, Mars, imaged by HiRISE. Credit: NASA/JPL-Caltech/University of Arizona

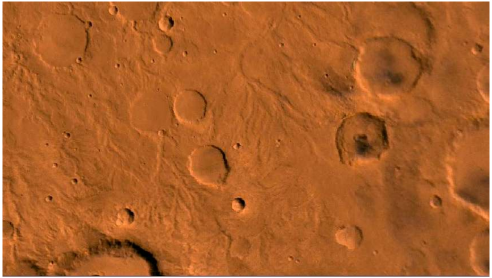


Barchan dunes in Nili Patera, Mars. Repeat HiRISE images show whole dunes advancing at sand fluxes comparable to Earth's, under an atmosphere 170 times thinner (Bridges et al. 2012). Credit: NASA/JPL-Caltech/University of Arizona.



Longitudinal dunes in Titan’s Shangri-La region, Cassini SAR: organic sand, water-ice bedrock, the same aeolian physics. Credit: NASA/JPL-Caltech/ASI.

Fluvial (water) processes

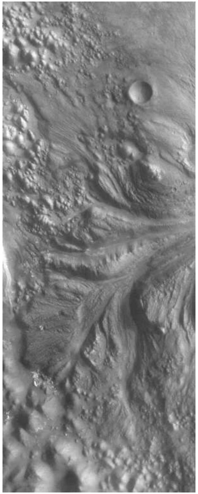


Credit: NASA/JPL/USGS

Mars valley networks: Noachian (>3.7 Ga), dendritic, sustained liquid water

Outflow channels: hundreds of km long, catastrophic groundwater release

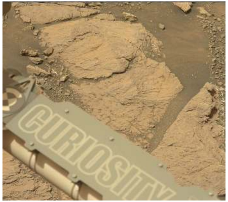
Titan: methane rivers; Huygens landed on rounded ice pebbles in a dry riverbed (2005)



An outflow channel through Aram Chaos: braided islands and scoured troughs record catastrophic floods, in contrast to the dendritic valley networks that record precipitation. Credit: NASA/JPL-Caltech/MSSS.

Glacial and chemical weathering

Fedchenko Glacier, Pamir. Credit: NASA/ISS



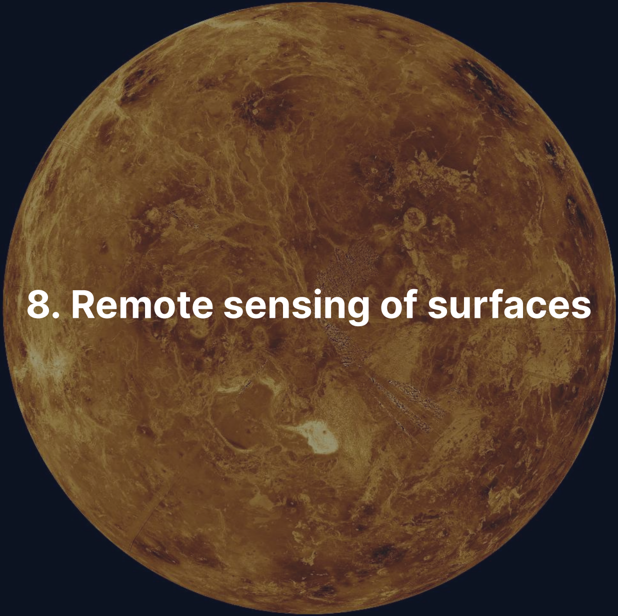
Clay-bearing unit, Glen Torridon, Gale crater. Credit:

NASA/JPL-Caltech/MSSS

Glacial: ice flows under its own weight. Earth's ice sheets covered ~30% of land at the Last Glacial Maximum; Mars has polar caps and buried mid-latitude ice.

Chemical: silicate weathering is Earth's carbon sink. On Mars, clays, sulfates and carbonates seen from orbit record past liquid water.

8. Remote sensing of surfaces



Venus in Magellan radar, the surface seen through an opaque atmosphere. Credit: NASA/JPL

Four ways to see a surface from orbit

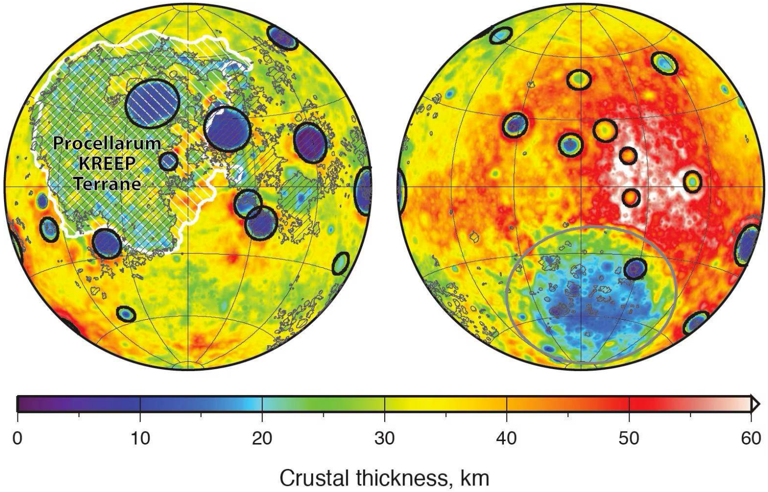
We land at a few sites. Almost all surface knowledge comes from orbit.

Reflectance spectroscopy minerals from absorption bands (CRISM at Mars: clays, carbonates)

Radar imaging through clouds and haze (Magellan at Venus, Cassini at Titan)

Laser altimetry topography to ~1 m (MOLA made the Mars map you just saw)

Gravity field mapping buried mass and crustal thickness (GRAIL at the Moon)



Lunar crustal thickness from GRAIL gravity and topography, nearside (left) and farside (right): mean 34–43 km, thin under the big basins, so a basin is a hole punched through most of the crust. Credit: NASA/MIT/GSFC/JPL-Caltech.

9. Regolith and space weathering

Apollo 11 bootprint in the lunar regolith. Credit: NASA

A shattered skin on every airless world

Credit: NASA/Apollo 11

Regolith: loose debris made by impact gardening; 2–4 m on the maria, 6–8 m on the highlands

Space weathering: solar wind and micrometeorites make nanophase iron; surfaces turn **darker and redder**, so fresh craters like Tycho look bright

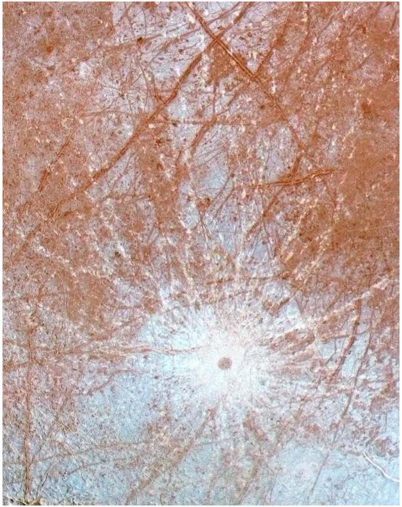
10. Cryovolcanism on icy bodies

Enceladus plumes. Credit: NASA/JPL-Caltech/SSI

Cryovolcanism: volcanism on ice worlds

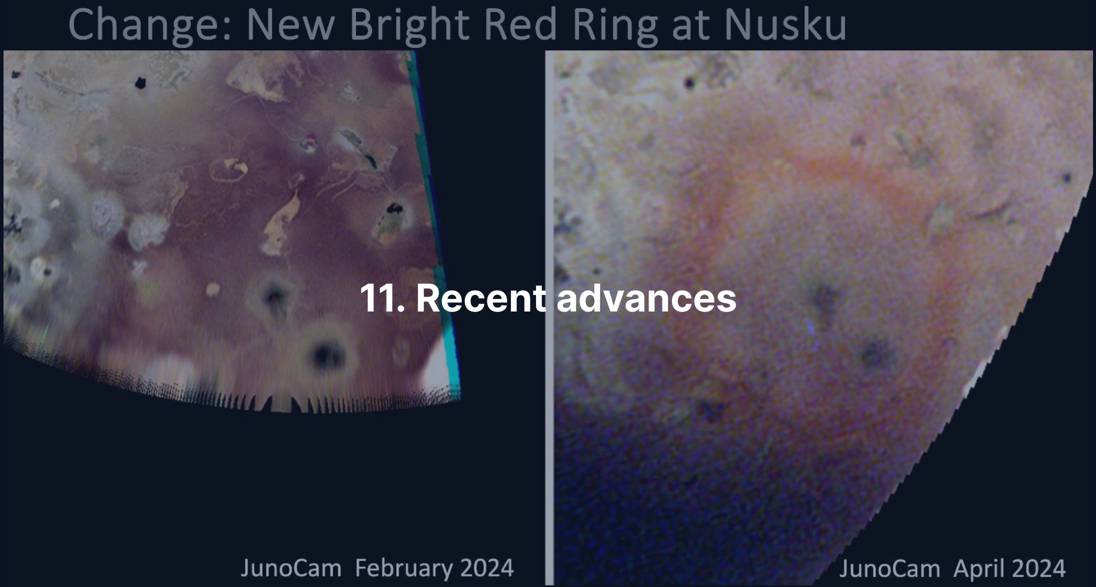
- The "magma" is water, ammonia-water or methane at 100–270 K; on Triton, nitrogen gas at ~38 K
- **Tidal heating** keeps oceans liquid under the ice shells
- Enceladus erupts water plumes, Europa resurfaces, Triton vents nitrogen

Liquid water far outside the habitable zone, kept by tides, not sunlight. The bodies return in Lecture 11.

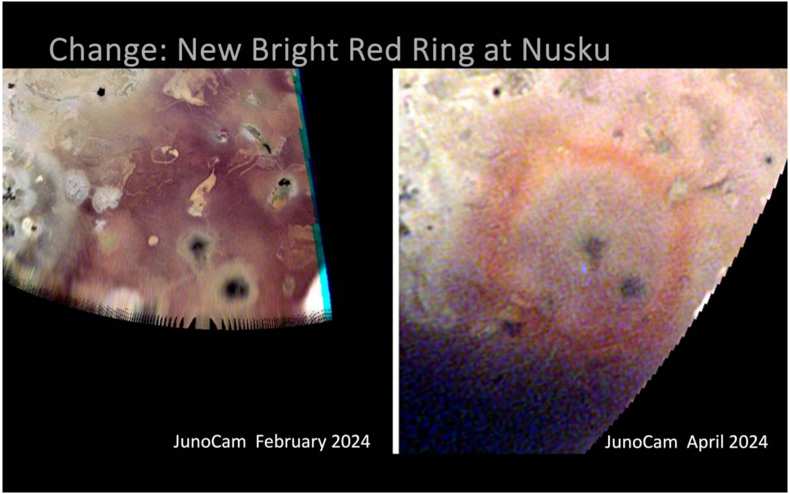


Double ridges and lineae near Pwyll crater on Europa, Galileo SSI. So few craters that the surface is only 40–90 Myr old. Credit: NASA/JPL-Caltech/University of Arizona/University of Colorado.

11. Recent advances



A new eruption on Io, JunoCam 2024. Credit: NASA/JPL-Caltech/SwRI/MSSS/Jason Perry



The Nusku region on Io in February and April 2024, JunoCam: a red ring of fresh sulfur-rich pyroclastic deposits appears between two flybys, the only body beyond Earth where volcanic resurfacing shows up on that timescale. Credit: NASA/JPL-Caltech/SwRI/MSSS/Jason Perry.

Summary

1. Surfaces record **endogenic** (volcanism, tectonics) against **exogenic** (impacts, erosion) processes
2. Crater scaling law: $D \sim (E/\rho g)^{1/4}$, a fourth root of the energy
3. Crater counting plus Apollo calibration gives absolute surface ages
4. No plate tectonics: giant volcanoes, one rift, a shrinking Mercury; Earth alone breaks its lid
5. Fluvial features on Mars are strong evidence for past liquid water

Next: Lecture 8 goes underneath, to the interiors that drive the endogenic half of this record.

Questions?

Next: Lecture 8, Planetary interiors: structure, composition & dynamics

Thu 1 Oct, 11:00–13:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 2 Oct, 13:00–15:00, room 5419.0161 (Collegezaal)

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